Category-Aware Location Embedding for Point-of-Interest Recommendation

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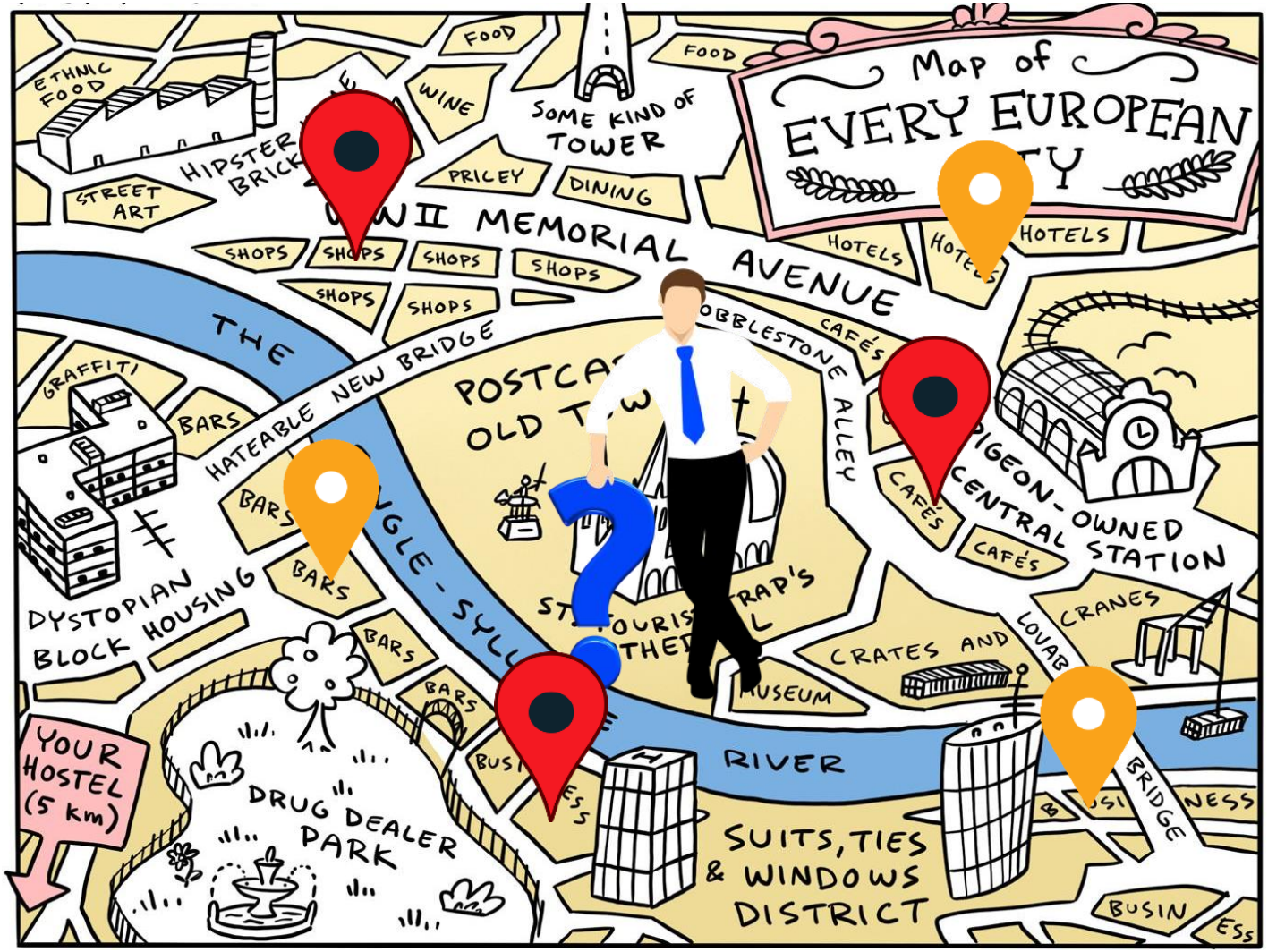
Map of
EVERY EUROPEAN
FOOD
FOOD
ETHNIC FOOD
WINE
SOME KIND OF TOWER
HIPSTER
STREET ART
PRICEY
DINING
WWII MEMORIAL AVENUE
SHOPS
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THE
HATEABLE NEW BRIDGE
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PIGEON-OWNED CENTRAL STATION
CAFÉS
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BARS
DYSTOPIAN BLOCK HOUSING
ST. TOURIS
THE
TRAP'S
MUSEUM
CRATES AND
CRANES
LOUAB
YOUR HOSTEL (5 km)
DRUG DEALER PARK
RIVER
BRIDGE
SUITS, TIES & WINDOWS DISTRICT
BUSIN
ESS

Motivations

- With the advent of Location-based Social Networks, users can share check-in data using their mobile devices, together with contextual information
- Successful results of considering pre-training models
- Propose a POI embedding model, focusing on the contextual information
- Account for sequential and categorical information
- Examine the effect of contextual information in pre-training model of POI

Contextual Information

- Sequence in the visiting of POIs
- Categorical information of POIs
- Examine the effect of contextual information in a pre-trained model of POI

Proposed method

- We propose a category-aware POI embedding model called CATAPE
- It captures both the geographical and categorical information of POIs
- Check-in Module: Visited POIs by users are geographically influenced by each other

$$\phi_{ch}(l, S_l) = arg\ max \prod_{l \in L} [\prod_{l_w \in W(l,S_l)} p(l_w|l)] \tag{1}$$

Proposed method (Cont.)

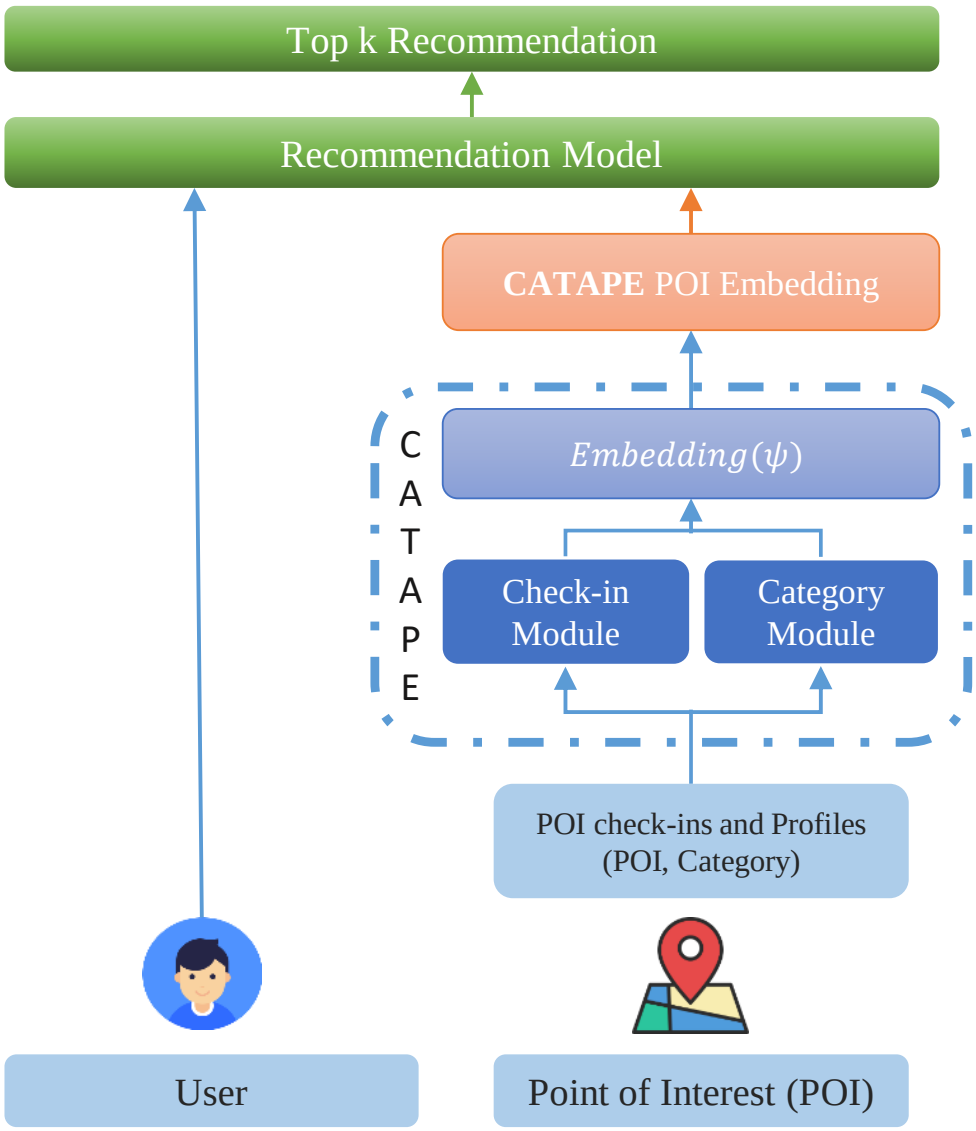
- Category Module: Categories of POIs visited show the user's interests.

$$\phi_{ct}(l, c) = arg\ max \prod_{l_i \in L} \prod_{c_j \in C_i} \prod_{c_q \in Q(c_j)} p(c_q | c_j, l_i) \tag{2}$$

- Objective Function

$$\psi_{emb} = \phi_{ch}(l, S_l) + \phi_{ct}(l, c) \tag{3}$$

Model workflow



Datasets

Datasets	#Users	#POIs	#Check-ins	#Categories
Yelp	6,254	21,240	357,410	507
Gowalla	10,162	24,250	456,988	417

Results on Yelp

Method	Yelp					
	P@5	P@10	P@20	R@5	R@10	R@20
USG	0.0282	0.0244	0.0197	0.0281	0.0523	0.0753
MGMPFM	0.0197	0.0173	0.0136	0.0211	0.0293	0.0493
BPRFM	0.0285	0.0221	0.0185	0.0296	0.0361	0.0599
RankGeoFM	0.0421	0.0362	0.0292	0.0392	0.0673	0.0838
HGMF	0.0532	0.0491	0.0401	0.0478	0.0702	0.0915
Metric Fact.	0.0593	0.0552	0.0481	0.0533	0.0782	0.0974
CATAPE-NoCat	0.0641†	0.0613†	0.0568†	0.0589	0.0831†	0.1013†
CATAPE	**0.0702**†‡	**0.0692**†‡	**0.0631**†‡	**0.0621**†‡	**0.0881**†	**0.1121**†‡

Results on Gowalla

Method	Gowalla					
	P@5	P@10	P@20	R@5	R@10	R@20
USG	0.0502	0.0471	0.0413	0.0517	0.0568	0.0625
MGMPFM	0.0281	0.0215	0.0197	0.0263	0.0291	0.0319
BPRFM	0.0493	0.0443	0.0342	0.0497	0.0529	0.0581
RankGeoFM	0.0567	0.0501	0.0492	0.0591	0.0642	0.0718
HGMF	0.0798	0.0711	0.0683	0.0715	0.0773	0.0819
Metric Fact.	0.0821	0.0782	0.0717	0.0784	0.0814	0.0862
CATAPE-NoCat	$0.0892^{\dagger}$	$0.0828^{\dagger}$	$0.0784^{\dagger}$	$0.0815^{\dagger}$	$0.0898^{\dagger}$	$0.0979^{\dagger}$
CATAPE	$\mathbf{0.0924}^{\dagger\ddagger}$	$\mathbf{0.0894}^{\dagger\ddagger}$	$\mathbf{0.0813}^{\dagger\ddagger}$	$\mathbf{0.0872}^{\dagger}$	$\mathbf{0.0953}^{\dagger\ddagger}$	$\mathbf{0.1283}^{\dagger\ddagger}$

Conclusions

- We proposed a novel POI embedding model: CATAPE
- CATAPE captures the sequential influence of POIs from check-in sequence of users
- CATAPE captures characteristics of POIs using the category information
- Our experiments show the effectiveness of CATAPE

Thanks for your attention!

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